

Circular colouring the orthogonality graph

OPG-2040

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $\mathcal{O}$ denote the graph with vertex set consisting of all lines through the origin in $\mathbb{R}^3$ and two vertices adjacent in $\mathcal{O}$ if they are perpendicular.

Problem Is $\chi_c(\mathcal{O}) = 4$?